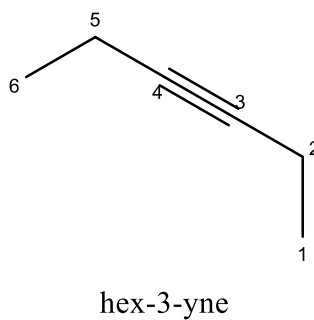
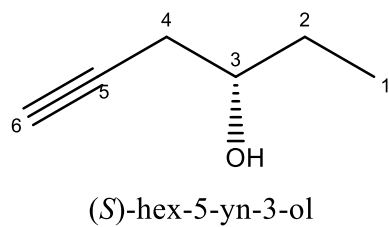
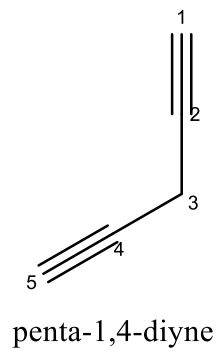
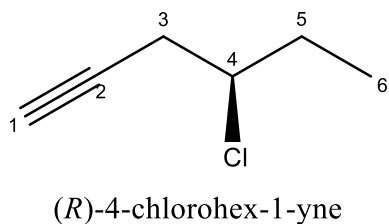


CH 221

KEY CH 9 Worksheet: Alkynes

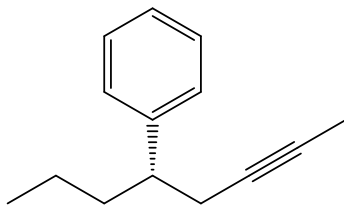
1. Provide the IUPAC name for the following compounds, including stereochemistry where necessary.



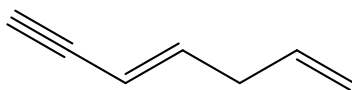
2. Draw the following molecules.

a. Acetylene: $\equiv$

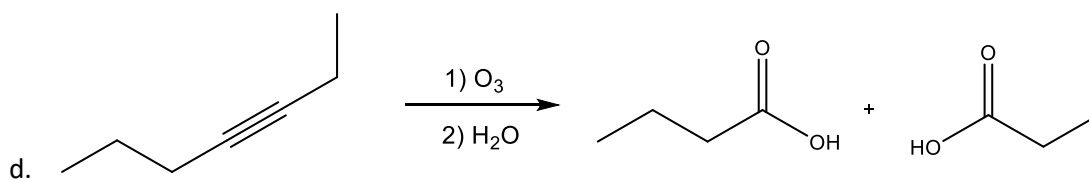
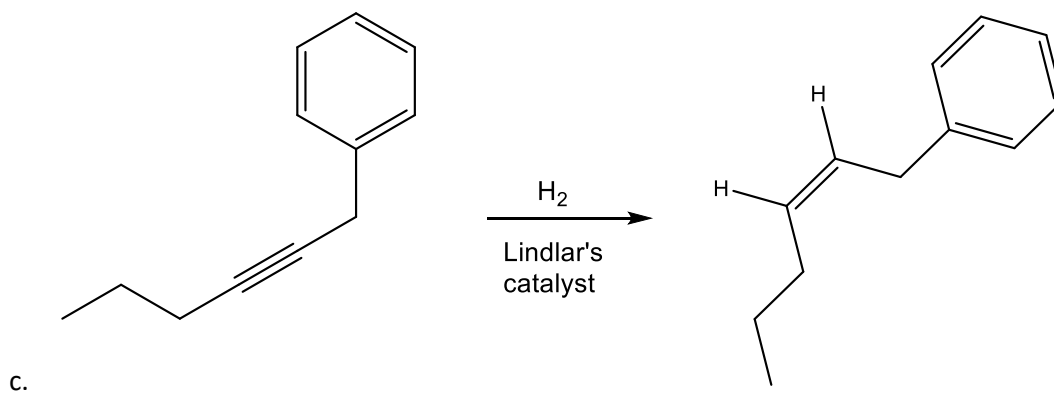
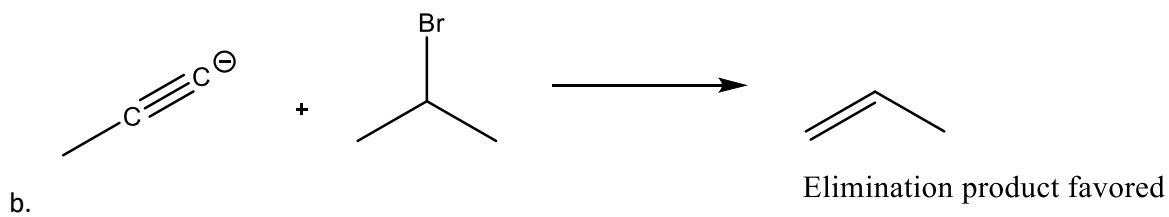
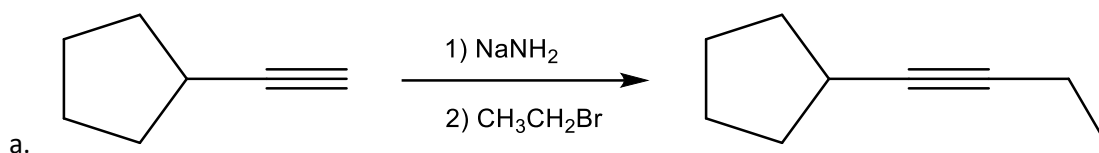
b. *(S)*-5-phenyloct-2-yne

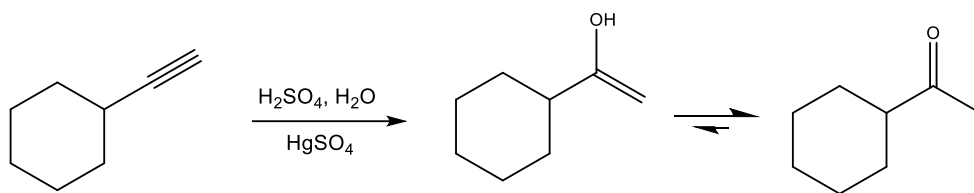


c. Hepta-3,6-dien-1-yne



3. Provide the structure of the major organic product(s) for the following reactions.

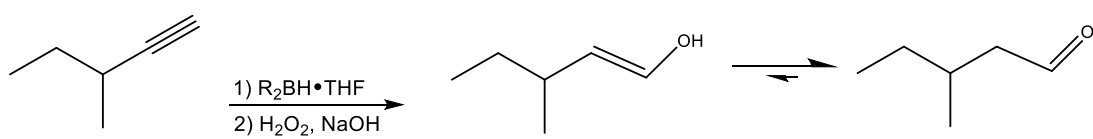




*Markovnikov Addition

Ketone major product after enol-keto tautomerization

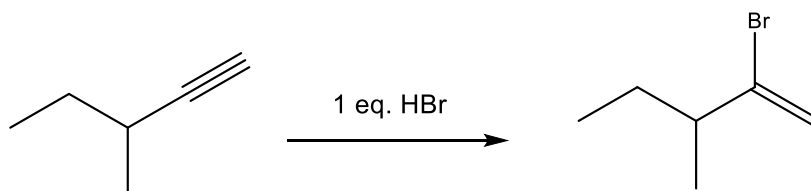
e.



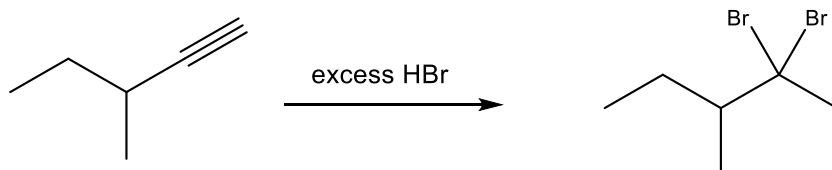
Anti-Markovnikov addition

Aldehyde is the favored tautomer

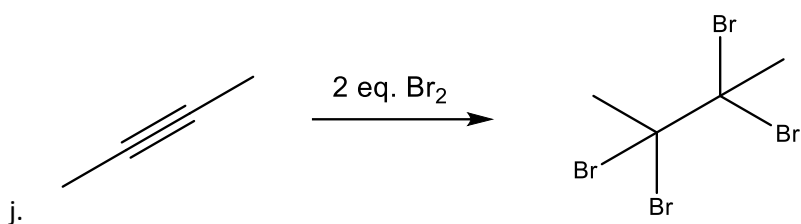
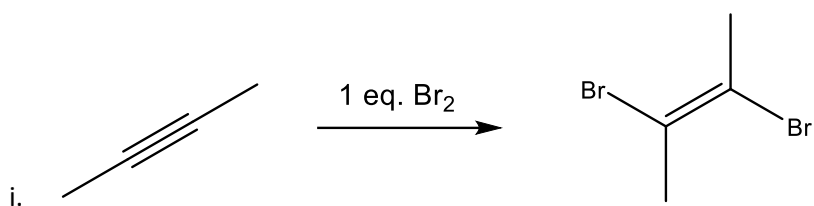
f.



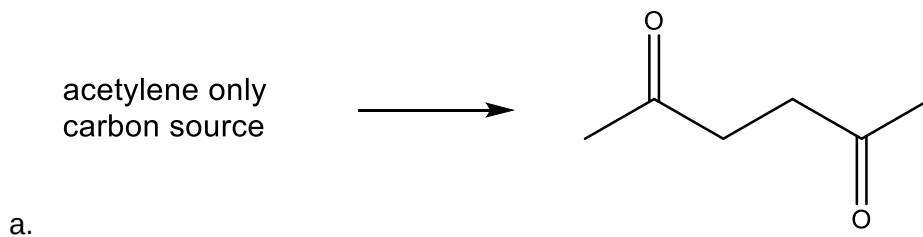
g.



h.



4. The following involve multistep synthesis. Propose a synthesis for the following:



b. Prepare racemic 2,3-dibromobutane from propyne.

c. Prepare hept-1-yne from hept-1-ene.

d. Prepare butylbenzene from phenylacetylene.

e. Prepare (Z)-pent-2-ene from propyne.

f. Prepare (E)-2,3-dibromobut-2-ene from butan-2-ol.

